Vishesh Prasad

US Citizen | +1 (972) 757-9367 | vprasad3@illinois.edu | https://www.visheshprasad.com

Research Interests

Statistical machine learning, sequential decision making, reinforcement learning, multi-agent systems, language models, mathematical reasoning, game theory.

Education

- **University of Illinois Urbana-Champaign** — *August 2025 - Expected: May 2027*
  M.S./Ph.D. Electrical & Computer Engineering — Urbana, IL
  - GPA: 4.00/4.00
- **University of Illinois Urbana-Champaign** — *August 2021 - May 2025*
  B.S. Computer Engineering — Urbana, IL
  - Highest Honors, GPA: 3.81/4.00
  - Minors: Mathematics, Statistics, Econometrics
  - Senior thesis: Extractive Summarization for Mathematically Rich Texts

Experience

- **Research Assistant** — *October 2025 - Present*
  CS at University of Illinois Urbana-Champaign
  - Researching adversarial reinforcement learning
  - Advisor: Dr. Arindam Banerjee
- **Research Assistant** — *August 2023 - Present*
  Mathematical Language Processing Group (MLP), ECE at University of Illinois Urbana-Champaign []
  - Exploring the use of diffusion models for mathematical reasoning
  - Designed novel algorithms and LLM techniques for mathematical relation extraction (Preprint **[M.1]**)
  - Advisor: Dr. Nickvash Kani
- **Research Assistant** — *August 2024 - May 2025*
  IBM-Illinois Discovery Accelerator Institute (IIDAI) []
  - Run-time and accuracy optimization of machine learning applications with a novel ML compiler
  - Advisors: Dr. Sasa Misailovic, Dr. Vikram Adve
- **Research Assistant** — *June 2024 - December 2024*
  Software Engineering and Analysis Lab (SEAL), CS at Cornell University
  - Machine Learning operations (MLOps) research study looking at MLflow, Metaflow, BentoML, ZenML, and Kubeflow
  - Advisor: Dr. Saikat Dutta
- **Software Engineering Intern** — *June 2024 - August 2024*
  GEICO — *Chevy Chase, MD*
  - Developed a knowledge graph using a Neo4j native graph database to analyze thousands of cloud and on-premises entities
  - Utilized the graph for machine learning predictions and cost optimization, resulting in potentially millions of dollars in savings
  - Built a Node.js application for performance-optimized database interaction and visualization
- **Embedded Software Developer Intern** — *May 2023 - December 2023*
  Lumentum — *Dallas, TX*
  - Added new features for a C# GUI by modifying device drivers to acquire greater controls of registers
  - Involved in setting up new IAR toolchains for ARM processors utilizing Jenkins and debugging embedded C code
  - Conducted field failure analysis support by replicating customer issues through microphonic tests

Teaching Experience

- **Graduate Teaching Assistant** — *August 2025 - Present*
  University of Illinois Department of Electrical and Computer Engineering
  - CS 446/ECE 449 – Machine Learning (Jan. 2026 - Present)
  - CS/ECE 374 – Algorithms & Models of Computation (Aug. 2025 - Dec. 2025)
- **Undergraduate Teaching Assistant** — *August 2022 - May 2025*
  University of Illinois Department of Electrical and Computer Engineering
  - ECE 364 – Programming Methods for Machine Learning (Jan. 2025 - May 2025)
  - CS/ECE 374 – Algorithms & Models of Computation (Jan. 2024 - Dec. 2024)
  - Engineering Economics – Course Development as Head TA (Apr. 2024 - July 2024)
  - ECE 110 – Introduction to Electronics (Aug. 2022 - Aug. 2023)

Relevant Coursework

- **Graduate Coursework:** Statistical Reinforcement Learning, Sequential Decision Making, Statistical Inference, Statistical Learning Theory, Diffusion Flow Matching Models, Random Processes, Algorithms, Algorithmic Market Microstructure
- **Upper-level Coursework:** Machine Learning, Stochastic Processes, Real Analysis, Financial Econometrics, Optimization, Economic Game Theory, Probabilistic Graph Models, Parallel Programming, Applied Econometrics, Artificial Intelligence, Digital Signal Processing, Computer Systems, Logic Synthesis, Advanced Competitive Algorithm Programming

Honors and Awards

- **Edward C. Jordan Award** — *April 2025*
  Department of Electrical and Computer Engineering at Illinois []
- **Indira Gunda Saladi Research Scholarship** — *April 2025*
  Department of Electrical and Computer Engineering at Illinois []
- **Illinois Scholars Undergraduate Research Scholarship** — *May 2024*
  Illinois Scholars Undergraduate Research (ISUR) Program []
- **Edmund J. James Scholar** — *August 2021 - May 2025*
  Department of Electrical and Computer Engineering at Illinois
- **Dean's List**
  University of Illinois Urbana-Champaign

Publications

M=Manuscript(Pre-Print)

[M.1] Prasad, et al. (2026). **Mathematical Derivation Graphs: A Relation Extraction Task in STEM Manuscripts**. In *arXiv*.

Student Organizations

- **Association for Quantitative Trading Education (AQTE)**
  - Education Director (April 2022 - January 2023)
- **EntreCOPRS - Startup Consulting**
  - Technology Director (May 2023 - December 2023)
  - Strategy Manager (December 2022 - May 2023)
  - Senior Consultant (August 2022 - December 2022)
  - Consultant (February 2022 - August 2022)

Additional Information

Languages: English (Fluent), Kannada (Fluent)
Interests: Chess, Hiking, Tenor Saxophone, Reading, Woodworking, LEGO, Football, Cricket